

Wave 1 Statistical Analyses

Marta Topor & Bertil Wegmann

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Hierarchical Bayesian Models Wave 1

This document runs the models used to analyse the main outcomes for Wave 1.
The structure of the document follows the Analysis Plans.txt file.
The variables and outcome variables are outlined in Meta-Data.txt.

Data Set 1 - Binary Categorical Outcomes

Crisis aware

Bayesian Logistic regression for whether participants were aware of the reproducibility crisis or not (yes/no).
The explainer variables are:

- Whether the participant was at a university that was affiliated with the UKRN - whether the participant was in their first, second, placement or third year

Bayesian model for UKRN/Non-UKRN:

```
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
## Chain 1:
## Chain 1: Gradient evaluation took 0.000359 seconds
## Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 3.59 seconds.
## Chain 1: Adjust your expectations accordingly!
## Chain 1:
## Chain 1:
## Chain 1: Iteration:    1 / 2000 [  0%] (Warmup)
## Chain 1: Iteration:   200 / 2000 [ 10%] (Warmup)
## Chain 1: Iteration:   400 / 2000 [ 20%] (Warmup)
## Chain 1: Iteration:   600 / 2000 [ 30%] (Warmup)
## Chain 1: Iteration:   800 / 2000 [ 40%] (Warmup)
## Chain 1: Iteration:  1000 / 2000 [ 50%] (Warmup)
## Chain 1: Iteration:  1001 / 2000 [ 50%] (Sampling)
## Chain 1: Iteration:  1200 / 2000 [ 60%] (Sampling)
## Chain 1: Iteration:  1400 / 2000 [ 70%] (Sampling)
## Chain 1: Iteration:  1600 / 2000 [ 80%] (Sampling)
## Chain 1: Iteration:  1800 / 2000 [ 90%] (Sampling)
## Chain 1: Iteration:  2000 / 2000 [100%] (Sampling)
## Chain 1:
## Chain 1: Elapsed Time: 2.264 seconds (Warm-up)
## Chain 1:                1.446 seconds (Sampling)
## Chain 1:                3.71 seconds (Total)
##
```

```

## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
## Chain 2:
## Chain 2: Gradient evaluation took 0.0001 seconds
## Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 1 seconds.
## Chain 2: Adjust your expectations accordingly!
## Chain 2:
## Chain 2:
## Chain 2: Iteration:    1 / 2000 [  0%] (Warmup)
## Chain 2: Iteration:   200 / 2000 [ 10%] (Warmup)
## Chain 2: Iteration:   400 / 2000 [ 20%] (Warmup)
## Chain 2: Iteration:   600 / 2000 [ 30%] (Warmup)
## Chain 2: Iteration:   800 / 2000 [ 40%] (Warmup)
## Chain 2: Iteration:  1000 / 2000 [ 50%] (Warmup)
## Chain 2: Iteration:  1001 / 2000 [ 50%] (Sampling)
## Chain 2: Iteration:  1200 / 2000 [ 60%] (Sampling)
## Chain 2: Iteration:  1400 / 2000 [ 70%] (Sampling)
## Chain 2: Iteration:  1600 / 2000 [ 80%] (Sampling)
## Chain 2: Iteration:  1800 / 2000 [ 90%] (Sampling)
## Chain 2: Iteration:  2000 / 2000 [100%] (Sampling)
## Chain 2:
## Chain 2: Elapsed Time: 1.426 seconds (Warm-up)
## Chain 2:                1.686 seconds (Sampling)
## Chain 2:                3.112 seconds (Total)
## Chain 2:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).
## Chain 3:
## Chain 3: Gradient evaluation took 0.000449 seconds
## Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 4.49 seconds.
## Chain 3: Adjust your expectations accordingly!
## Chain 3:
## Chain 3:
## Chain 3: Iteration:    1 / 2000 [  0%] (Warmup)
## Chain 3: Iteration:   200 / 2000 [ 10%] (Warmup)
## Chain 3: Iteration:   400 / 2000 [ 20%] (Warmup)
## Chain 3: Iteration:   600 / 2000 [ 30%] (Warmup)
## Chain 3: Iteration:   800 / 2000 [ 40%] (Warmup)
## Chain 3: Iteration:  1000 / 2000 [ 50%] (Warmup)
## Chain 3: Iteration:  1001 / 2000 [ 50%] (Sampling)
## Chain 3: Iteration:  1200 / 2000 [ 60%] (Sampling)
## Chain 3: Iteration:  1400 / 2000 [ 70%] (Sampling)
## Chain 3: Iteration:  1600 / 2000 [ 80%] (Sampling)
## Chain 3: Iteration:  1800 / 2000 [ 90%] (Sampling)
## Chain 3: Iteration:  2000 / 2000 [100%] (Sampling)
## Chain 3:
## Chain 3: Elapsed Time: 1.71 seconds (Warm-up)
## Chain 3:                1.638 seconds (Sampling)
## Chain 3:                3.348 seconds (Total)
## Chain 3:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).
## Chain 4:
## Chain 4: Gradient evaluation took 0.000102 seconds
## Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 1.02 seconds.

```

```

## Chain 4: Adjust your expectations accordingly!
## Chain 4:
## Chain 4:
## Chain 4: Iteration:    1 / 2000 [ 0%] (Warmup)
## Chain 4: Iteration:   200 / 2000 [ 10%] (Warmup)
## Chain 4: Iteration:   400 / 2000 [ 20%] (Warmup)
## Chain 4: Iteration:   600 / 2000 [ 30%] (Warmup)
## Chain 4: Iteration:   800 / 2000 [ 40%] (Warmup)
## Chain 4: Iteration:  1000 / 2000 [ 50%] (Warmup)
## Chain 4: Iteration:  1001 / 2000 [ 50%] (Sampling)
## Chain 4: Iteration:  1200 / 2000 [ 60%] (Sampling)
## Chain 4: Iteration:  1400 / 2000 [ 70%] (Sampling)
## Chain 4: Iteration:  1600 / 2000 [ 80%] (Sampling)
## Chain 4: Iteration:  1800 / 2000 [ 90%] (Sampling)
## Chain 4: Iteration:  2000 / 2000 [100%] (Sampling)
## Chain 4:
## Chain 4: Elapsed Time: 1.717 seconds (Warm-up)
## Chain 4:           2.223 seconds (Sampling)
## Chain 4:           3.94 seconds (Total)
## Chain 4:

## Inference for Stan model: anon_model.
## 4 chains, each with iter=2000; warmup=1000; thin=1;
## post-warmup draws per chain=1000, total post-warmup draws=4000.
##
##           mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu          0.24    0.00  0.17   -0.10    0.13    0.24    0.35    0.55  1673
## sigma        0.51    0.00  0.14    0.28    0.42    0.51    0.59    0.82  1442
## alpha[1]     0.66    0.00  0.32    0.06    0.44    0.65    0.86    1.30  4134
## alpha[2]     0.50    0.00  0.30   -0.09    0.30    0.50    0.70    1.12  4471
## alpha[3]     0.17    0.01  0.33   -0.50   -0.05    0.18    0.40    0.81  2854
## alpha[4]    -0.44    0.01  0.37   -1.18   -0.69   -0.43   -0.19    0.26  1824
## alpha[5]     0.34    0.01  0.34   -0.34    0.11    0.34    0.56    1.03  3031
## alpha[6]    -0.09    0.01  0.34   -0.76   -0.32   -0.08    0.13    0.58  2134
## alpha[7]    -0.36    0.01  0.36   -1.10   -0.61   -0.36   -0.11    0.32  1726
## alpha[8]    -0.18    0.01  0.34   -0.84   -0.40   -0.17    0.05    0.47  1904
## alpha[9]     0.26    0.01  0.31   -0.35    0.05    0.27    0.48    0.86  3422
## alpha[10]   -0.09    0.01  0.30   -0.68   -0.29   -0.09    0.11    0.50  2498
## alpha[11]   -0.02    0.00  0.30   -0.62   -0.21   -0.01    0.19    0.54  3978
## alpha[12]   -0.12    0.01  0.31   -0.73   -0.32   -0.11    0.10    0.47  3056
## alpha[13]    0.08    0.01  0.30   -0.53   -0.12    0.09    0.28    0.65  3501
## alpha[14]    0.17    0.00  0.30   -0.44   -0.03    0.17    0.37    0.74  4222
## alpha[15]    0.61    0.00  0.31    0.01    0.40    0.61    0.82    1.24  4178
## alpha[16]    0.57    0.00  0.29    0.00    0.37    0.56    0.76    1.15  3894
## alpha[17]    0.24    0.00  0.29   -0.34    0.05    0.24    0.44    0.81  4123
## alpha[18]   -0.14    0.01  0.33   -0.79   -0.35   -0.14    0.09    0.49  2307
## alpha[19]    0.87    0.01  0.36    0.22    0.62    0.85    1.10    1.62  4062
## alpha[20]    0.44    0.00  0.30   -0.17    0.24    0.43    0.63    1.07  4470
## alpha[21]    0.87    0.01  0.34    0.25    0.63    0.85    1.09    1.58  3634
## alpha[22]    0.06    0.00  0.28   -0.49   -0.13    0.06    0.25    0.61  4347
## alpha[23]    1.00    0.01  0.35    0.36    0.75    0.98    1.23    1.73  3076
## beta         0.54    0.01  0.20    0.16    0.41    0.54    0.67    0.94  1244
## lp__        -510.53   0.12  4.27  -519.47  -513.25  -510.32  -507.62  -502.70  1201
##           Rhat

```

```

## mu          1
## sigma       1
## alpha[1]    1
## alpha[2]    1
## alpha[3]    1
## alpha[4]    1
## alpha[5]    1
## alpha[6]    1
## alpha[7]    1
## alpha[8]    1
## alpha[9]    1
## alpha[10]   1
## alpha[11]   1
## alpha[12]   1
## alpha[13]   1
## alpha[14]   1
## alpha[15]   1
## alpha[16]   1
## alpha[17]   1
## alpha[18]   1
## alpha[19]   1
## alpha[20]   1
## alpha[21]   1
## alpha[22]   1
## alpha[23]   1
## beta        1
## lp__        1
##
## Samples were drawn using NUTS(diag_e) at Mon Jun 17 12:40:01 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

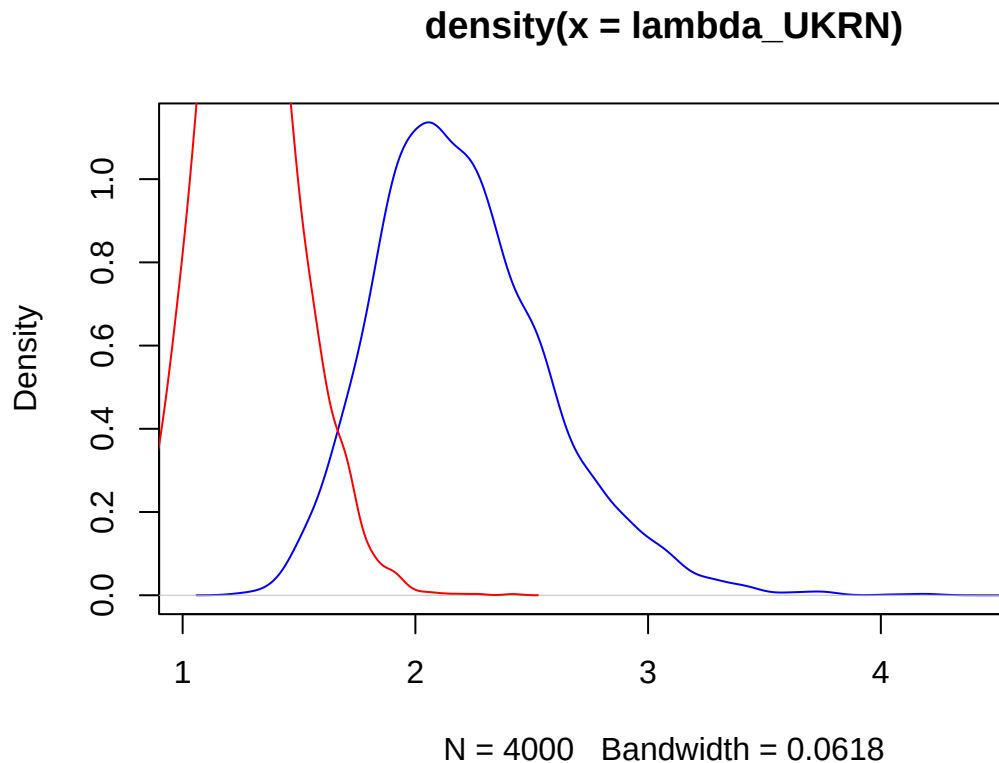
```

The output shows that beta credibility intervals cross 0.

2.5% = -0.19

97.5% = 2.13

So the interpretation is that students who attend a UKRN affiliated university are not more likely to be aware of the reproducibility crisis.



To see the magnitude of this difference.

```
## [1] 0.99825
```

Mean lambda for students in Non-UKRN university = 1.33

Mean lambda for students in a UKRN university = 3.56

Difference = 0.95

Bayesian model for differences in year groups.

Learnt Open Science at Uni

Bayesian Logistic regression for whether participants received teaching about open science or not (yes/no).

The explainer variables are:

- Whether the participant was at a university that was affiliated with the UKRN - whether the participant was in their first, second, placement or third year

Bayesian model for UKRN/Non-UKRN:

```
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
## Chain 1:
## Chain 1: Gradient evaluation took 0.000237 seconds
## Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 2.37 seconds.
## Chain 1: Adjust your expectations accordingly!
## Chain 1:
## Chain 1:
## Chain 1: Iteration:    1 / 2000 [  0%] (Warmup)
## Chain 1: Iteration:  200 / 2000 [ 10%] (Warmup)
```

```

## Chain 1: Iteration: 400 / 2000 [ 20%] (Warmup)
## Chain 1: Iteration: 600 / 2000 [ 30%] (Warmup)
## Chain 1: Iteration: 800 / 2000 [ 40%] (Warmup)
## Chain 1: Iteration: 1000 / 2000 [ 50%] (Warmup)
## Chain 1: Iteration: 1001 / 2000 [ 50%] (Sampling)
## Chain 1: Iteration: 1200 / 2000 [ 60%] (Sampling)
## Chain 1: Iteration: 1400 / 2000 [ 70%] (Sampling)
## Chain 1: Iteration: 1600 / 2000 [ 80%] (Sampling)
## Chain 1: Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 1: Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 1:
## Chain 1: Elapsed Time: 1.698 seconds (Warm-up)
## Chain 1: 2.201 seconds (Sampling)
## Chain 1: 3.899 seconds (Total)
## Chain 1:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
## Chain 2:
## Chain 2: Gradient evaluation took 0.000185 seconds
## Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 1.85 seconds.
## Chain 2: Adjust your expectations accordingly!
## Chain 2:
## Chain 2:
## Chain 2: Iteration: 1 / 2000 [ 0%] (Warmup)
## Chain 2: Iteration: 200 / 2000 [ 10%] (Warmup)
## Chain 2: Iteration: 400 / 2000 [ 20%] (Warmup)
## Chain 2: Iteration: 600 / 2000 [ 30%] (Warmup)
## Chain 2: Iteration: 800 / 2000 [ 40%] (Warmup)
## Chain 2: Iteration: 1000 / 2000 [ 50%] (Warmup)
## Chain 2: Iteration: 1001 / 2000 [ 50%] (Sampling)
## Chain 2: Iteration: 1200 / 2000 [ 60%] (Sampling)
## Chain 2: Iteration: 1400 / 2000 [ 70%] (Sampling)
## Chain 2: Iteration: 1600 / 2000 [ 80%] (Sampling)
## Chain 2: Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 2: Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 2:
## Chain 2: Elapsed Time: 1.918 seconds (Warm-up)
## Chain 2: 1.617 seconds (Sampling)
## Chain 2: 3.535 seconds (Total)
## Chain 2:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).
## Chain 3:
## Chain 3: Gradient evaluation took 0.000198 seconds
## Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 1.98 seconds.
## Chain 3: Adjust your expectations accordingly!
## Chain 3:
## Chain 3:
## Chain 3: Iteration: 1 / 2000 [ 0%] (Warmup)
## Chain 3: Iteration: 200 / 2000 [ 10%] (Warmup)
## Chain 3: Iteration: 400 / 2000 [ 20%] (Warmup)
## Chain 3: Iteration: 600 / 2000 [ 30%] (Warmup)
## Chain 3: Iteration: 800 / 2000 [ 40%] (Warmup)
## Chain 3: Iteration: 1000 / 2000 [ 50%] (Warmup)

```

```

## Chain 3: Iteration: 1001 / 2000 [ 50%] (Sampling)
## Chain 3: Iteration: 1200 / 2000 [ 60%] (Sampling)
## Chain 3: Iteration: 1400 / 2000 [ 70%] (Sampling)
## Chain 3: Iteration: 1600 / 2000 [ 80%] (Sampling)
## Chain 3: Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 3: Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 3:
## Chain 3: Elapsed Time: 2.486 seconds (Warm-up)
## Chain 3: 2.189 seconds (Sampling)
## Chain 3: 4.675 seconds (Total)
## Chain 3:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).
## Chain 4:
## Chain 4: Gradient evaluation took 8.8e-05 seconds
## Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.88 seconds.
## Chain 4: Adjust your expectations accordingly!
## Chain 4:
## Chain 4:
## Chain 4: Iteration: 1 / 2000 [ 0%] (Warmup)
## Chain 4: Iteration: 200 / 2000 [ 10%] (Warmup)
## Chain 4: Iteration: 400 / 2000 [ 20%] (Warmup)
## Chain 4: Iteration: 600 / 2000 [ 30%] (Warmup)
## Chain 4: Iteration: 800 / 2000 [ 40%] (Warmup)
## Chain 4: Iteration: 1000 / 2000 [ 50%] (Warmup)
## Chain 4: Iteration: 1001 / 2000 [ 50%] (Sampling)
## Chain 4: Iteration: 1200 / 2000 [ 60%] (Sampling)
## Chain 4: Iteration: 1400 / 2000 [ 70%] (Sampling)
## Chain 4: Iteration: 1600 / 2000 [ 80%] (Sampling)
## Chain 4: Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 4: Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 4:
## Chain 4: Elapsed Time: 1.433 seconds (Warm-up)
## Chain 4: 1.506 seconds (Sampling)
## Chain 4: 2.939 seconds (Total)
## Chain 4:

## Warning: There were 19 divergent transitions after warmup. See
## https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
## to find out why this is a problem and how to eliminate them.

## Warning: Examine the pairs() plot to diagnose sampling problems

## Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and medians may be
## Running the chains for more iterations may help. See
## https://mc-stan.org/misc/warnings.html#bulk-ess

## Warning: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and tail quant.
## Running the chains for more iterations may help. See
## https://mc-stan.org/misc/warnings.html#tail-ess

## Inference for Stan model: anon_model.
## 4 chains, each with iter=2000; warmup=1000; thin=1;
## post-warmup draws per chain=1000, total post-warmup draws=4000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff

```

```

## mu          -1.80    0.01 0.18   -2.17   -1.91   -1.79   -1.68   -1.48   706
## sigma        0.38    0.01 0.14    0.13    0.28    0.37    0.47    0.67   479
## alpha[1]     -1.54    0.01 0.30   -2.12   -1.75   -1.54   -1.35   -0.92  1936
## alpha[2]     -1.71    0.01 0.31   -2.34   -1.91   -1.70   -1.51   -1.12  2731
## alpha[3]     -2.00    0.01 0.35   -2.73   -2.22   -1.99   -1.75   -1.39  1071
## alpha[4]     -2.26    0.01 0.40   -3.14   -2.51   -2.22   -1.97   -1.59   771
## alpha[5]     -1.87    0.01 0.32   -2.54   -2.08   -1.85   -1.65   -1.28  1380
## alpha[6]     -2.18    0.01 0.39   -3.03   -2.43   -2.15   -1.90   -1.52   844
## alpha[7]     -2.09    0.01 0.37   -2.89   -2.32   -2.06   -1.83   -1.45   991
## alpha[8]     -1.73    0.01 0.31   -2.37   -1.93   -1.72   -1.51   -1.11  1842
## alpha[9]     -1.89    0.01 0.33   -2.59   -2.10   -1.86   -1.66   -1.31  1473
## alpha[10]    -2.00    0.01 0.35   -2.73   -2.22   -1.98   -1.76   -1.36  1315
## alpha[11]    -1.73    0.01 0.32   -2.39   -1.93   -1.72   -1.52   -1.12  3087
## alpha[12]    -1.82    0.01 0.31   -2.45   -2.02   -1.80   -1.61   -1.22  2012
## alpha[13]    -1.86    0.01 0.32   -2.55   -2.05   -1.85   -1.65   -1.26  1780
## alpha[14]    -1.82    0.01 0.32   -2.47   -2.02   -1.80   -1.61   -1.23  1904
## alpha[15]    -1.79    0.01 0.31   -2.41   -1.99   -1.77   -1.59   -1.20  2520
## alpha[16]    -1.85    0.01 0.32   -2.55   -2.05   -1.83   -1.64   -1.26  2076
## alpha[17]    -1.65    0.01 0.30   -2.26   -1.84   -1.65   -1.46   -1.05  2483
## alpha[18]    -1.89    0.01 0.32   -2.54   -2.09   -1.87   -1.66   -1.31  1344
## alpha[19]    -1.32    0.01 0.32   -1.90   -1.54   -1.33   -1.10   -0.68  1586
## alpha[20]    -1.66    0.01 0.30   -2.26   -1.85   -1.66   -1.47   -1.08  2479
## alpha[21]    -1.58    0.01 0.30   -2.17   -1.78   -1.59   -1.40   -0.97  2322
## alpha[22]    -1.95    0.01 0.33   -2.64   -2.15   -1.93   -1.72   -1.32  1926
## alpha[23]    -1.26    0.01 0.32   -1.83   -1.50   -1.28   -1.05   -0.59  1691
## beta          0.91    0.01 0.21    0.51    0.77    0.90    1.05    1.34   704
## lp__         -395.89   0.42 6.85 -408.12 -400.29 -396.39 -392.20 -379.37   272
##
## Rhat
## mu          1.00
## sigma       1.01
## alpha[1]    1.00
## alpha[2]    1.00
## alpha[3]    1.00
## alpha[4]    1.01
## alpha[5]    1.00
## alpha[6]    1.00
## alpha[7]    1.00
## alpha[8]    1.00
## alpha[9]    1.00
## alpha[10]   1.00
## alpha[11]   1.00
## alpha[12]   1.00
## alpha[13]   1.00
## alpha[14]   1.00
## alpha[15]   1.00
## alpha[16]   1.00
## alpha[17]   1.00
## alpha[18]   1.00
## alpha[19]   1.00
## alpha[20]   1.00
## alpha[21]   1.00
## alpha[22]   1.00
## alpha[23]   1.00
## beta        1.01

```

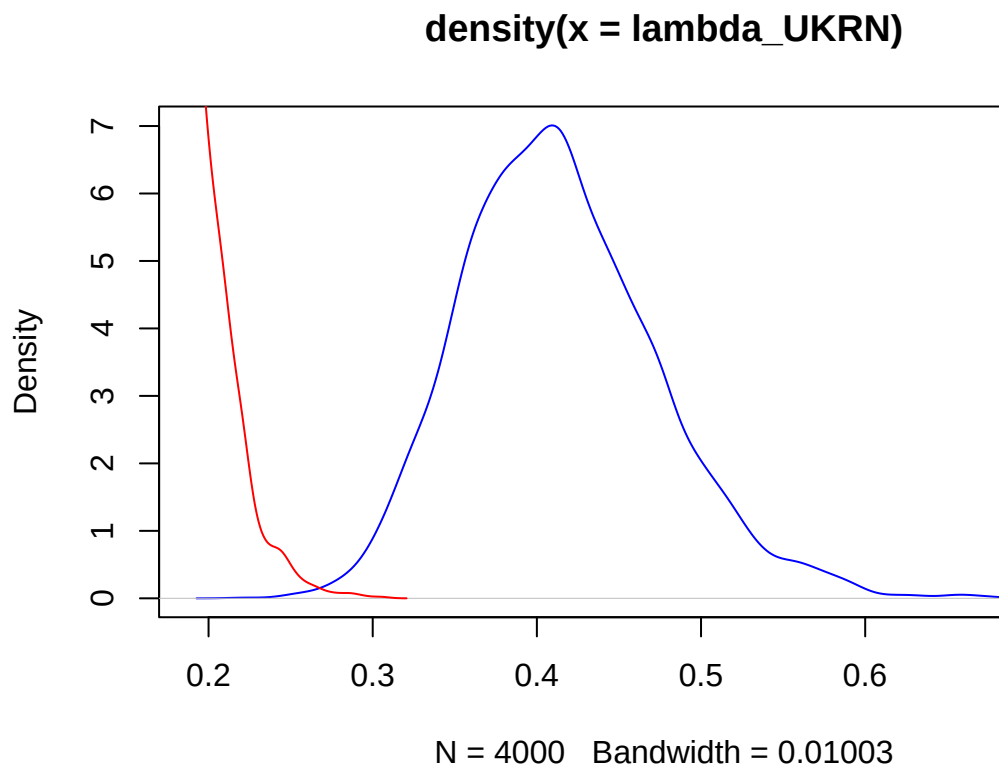
```
## lp__      1.02
##
## Samples were drawn using NUTS(diag_e) at Mon Jun 17 12:40:17 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

The output shows that beta credibility intervals do not cross 0.

2.5% = 0.14

97.5% = 1.90

So the interpretation is that students who attend a UKRN affiliated university are more likely to have learnt about open science.



To see the magnitude of this difference.

```
## [1] 1
```

Mean lambda for students in Non-UKRN university = 0.17

Mean lambda for students in a UKRN university = 0.48

Difference = 0.986

Can explain open science

Bayesian Logistic regression for whether participants can explain what open science is (yes/no). The explainer variables are:

- Whether the participant was at a university that was affiliated with the UKRN - whether the participant was in their first, second, placement or third year

Bayesian model for UKRN/Non-UKRN:

```

##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
## Chain 1:
## Chain 1: Gradient evaluation took 0.000297 seconds
## Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 2.97 seconds.
## Chain 1: Adjust your expectations accordingly!
## Chain 1:
## Chain 1:
## Chain 1: Iteration:    1 / 2000 [  0%] (Warmup)
## Chain 1: Iteration:   200 / 2000 [ 10%] (Warmup)
## Chain 1: Iteration:   400 / 2000 [ 20%] (Warmup)
## Chain 1: Iteration:   600 / 2000 [ 30%] (Warmup)
## Chain 1: Iteration:   800 / 2000 [ 40%] (Warmup)
## Chain 1: Iteration:  1000 / 2000 [ 50%] (Warmup)
## Chain 1: Iteration: 1001 / 2000 [ 50%] (Sampling)
## Chain 1: Iteration: 1200 / 2000 [ 60%] (Sampling)
## Chain 1: Iteration: 1400 / 2000 [ 70%] (Sampling)
## Chain 1: Iteration: 1600 / 2000 [ 80%] (Sampling)
## Chain 1: Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 1: Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 1:
## Chain 1: Elapsed Time: 2.17 seconds (Warm-up)
## Chain 1:                2.849 seconds (Sampling)
## Chain 1:                5.019 seconds (Total)
## Chain 1:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
## Chain 2:
## Chain 2: Gradient evaluation took 0.000203 seconds
## Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 2.03 seconds.
## Chain 2: Adjust your expectations accordingly!
## Chain 2:
## Chain 2:
## Chain 2: Iteration:    1 / 2000 [  0%] (Warmup)
## Chain 2: Iteration:   200 / 2000 [ 10%] (Warmup)
## Chain 2: Iteration:   400 / 2000 [ 20%] (Warmup)
## Chain 2: Iteration:   600 / 2000 [ 30%] (Warmup)
## Chain 2: Iteration:   800 / 2000 [ 40%] (Warmup)
## Chain 2: Iteration:  1000 / 2000 [ 50%] (Warmup)
## Chain 2: Iteration: 1001 / 2000 [ 50%] (Sampling)
## Chain 2: Iteration: 1200 / 2000 [ 60%] (Sampling)
## Chain 2: Iteration: 1400 / 2000 [ 70%] (Sampling)
## Chain 2: Iteration: 1600 / 2000 [ 80%] (Sampling)
## Chain 2: Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 2: Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 2:
## Chain 2: Elapsed Time: 2.747 seconds (Warm-up)
## Chain 2:                2.257 seconds (Sampling)
## Chain 2:                5.004 seconds (Total)
## Chain 2:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).
## Chain 3:
## Chain 3: Gradient evaluation took 9e-05 seconds

```

```

## Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.9 seconds.
## Chain 3: Adjust your expectations accordingly!
## Chain 3:
## Chain 3:
## Chain 3: Iteration:    1 / 2000 [  0%] (Warmup)
## Chain 3: Iteration:   200 / 2000 [ 10%] (Warmup)
## Chain 3: Iteration:   400 / 2000 [ 20%] (Warmup)
## Chain 3: Iteration:   600 / 2000 [ 30%] (Warmup)
## Chain 3: Iteration:   800 / 2000 [ 40%] (Warmup)
## Chain 3: Iteration:  1000 / 2000 [ 50%] (Warmup)
## Chain 3: Iteration: 1001 / 2000 [ 50%] (Sampling)
## Chain 3: Iteration: 1200 / 2000 [ 60%] (Sampling)
## Chain 3: Iteration: 1400 / 2000 [ 70%] (Sampling)
## Chain 3: Iteration: 1600 / 2000 [ 80%] (Sampling)
## Chain 3: Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 3: Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 3:
## Chain 3: Elapsed Time: 2.586 seconds (Warm-up)
## Chain 3:                1.401 seconds (Sampling)
## Chain 3:                3.987 seconds (Total)
## Chain 3:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).
## Chain 4:
## Chain 4: Gradient evaluation took 0.000228 seconds
## Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 2.28 seconds.
## Chain 4: Adjust your expectations accordingly!
## Chain 4:
## Chain 4:
## Chain 4: Iteration:    1 / 2000 [  0%] (Warmup)
## Chain 4: Iteration:   200 / 2000 [ 10%] (Warmup)
## Chain 4: Iteration:   400 / 2000 [ 20%] (Warmup)
## Chain 4: Iteration:   600 / 2000 [ 30%] (Warmup)
## Chain 4: Iteration:   800 / 2000 [ 40%] (Warmup)
## Chain 4: Iteration:  1000 / 2000 [ 50%] (Warmup)
## Chain 4: Iteration: 1001 / 2000 [ 50%] (Sampling)
## Chain 4: Iteration: 1200 / 2000 [ 60%] (Sampling)
## Chain 4: Iteration: 1400 / 2000 [ 70%] (Sampling)
## Chain 4: Iteration: 1600 / 2000 [ 80%] (Sampling)
## Chain 4: Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 4: Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 4:
## Chain 4: Elapsed Time: 2.485 seconds (Warm-up)
## Chain 4:                1.451 seconds (Sampling)
## Chain 4:                3.936 seconds (Total)
## Chain 4:

## Warning: There were 157 divergent transitions after warmup. See
## https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
## to find out why this is a problem and how to eliminate them.

## Warning: There were 3 chains where the estimated Bayesian Fraction of Missing Information was low. See
## https://mc-stan.org/misc/warnings.html#bfmi-low

## Warning: Examine the pairs() plot to diagnose sampling problems

```

```

## Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and medians may be
## Running the chains for more iterations may help. See
## https://mc-stan.org/misc/warnings.html#bulk-ess

## Warning: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and tail quant
## Running the chains for more iterations may help. See
## https://mc-stan.org/misc/warnings.html#tail-ess

## Inference for Stan model: anon_model.
## 4 chains, each with iter=2000; warmup=1000; thin=1;
## post-warmup draws per chain=1000, total post-warmup draws=4000.
##
##               mean se_mean    sd    2.5%    25%    50%    75%    97.5% n_eff
## mu            -1.20    0.01  0.12   -1.44   -1.27   -1.20   -1.12   -0.95   399
## sigma          0.20    0.01  0.12    0.05    0.11    0.19    0.28    0.47   182
## alpha[1]      -1.17    0.01  0.21   -1.60   -1.29   -1.17   -1.05   -0.72  1215
## alpha[2]      -1.21    0.01  0.21   -1.64   -1.33   -1.21   -1.08   -0.81  1263
## alpha[3]      -1.30    0.01  0.25   -1.85   -1.43   -1.26   -1.14   -0.87   604
## alpha[4]      -1.31    0.01  0.25   -1.89   -1.44   -1.27   -1.15   -0.89   693
## alpha[5]      -1.19    0.01  0.22   -1.66   -1.31   -1.19   -1.07   -0.75  1134
## alpha[6]      -1.27    0.01  0.23   -1.81   -1.40   -1.25   -1.13   -0.85   775
## alpha[7]      -1.33    0.01  0.26   -1.95   -1.47   -1.29   -1.17   -0.89   572
## alpha[8]      -1.13    0.01  0.22   -1.57   -1.26   -1.14   -1.00   -0.69   936
## alpha[9]      -1.24    0.01  0.22   -1.71   -1.36   -1.22   -1.11   -0.83   801
## alpha[10]     -1.30    0.01  0.23   -1.84   -1.41   -1.27   -1.15   -0.90   698
## alpha[11]     -1.16    0.01  0.20   -1.57   -1.29   -1.17   -1.04   -0.75  1188
## alpha[12]     -1.36    0.01  0.26   -2.01   -1.50   -1.31   -1.19   -0.95   465
## alpha[13]     -1.26    0.01  0.22   -1.78   -1.38   -1.24   -1.12   -0.88   801
## alpha[14]     -1.12    0.01  0.21   -1.52   -1.25   -1.14   -1.01   -0.67  1049
## alpha[15]     -1.17    0.01  0.21   -1.61   -1.29   -1.17   -1.04   -0.73  1395
## alpha[16]     -1.11    0.01  0.21   -1.49   -1.23   -1.13   -0.99   -0.66   912
## alpha[17]     -1.04    0.01  0.23   -1.42   -1.19   -1.08   -0.91   -0.53   613
## alpha[18]     -1.22    0.01  0.21   -1.71   -1.34   -1.22   -1.09   -0.82   905
## alpha[19]     -1.02    0.01  0.23   -1.41   -1.19   -1.06   -0.88   -0.51   441
## alpha[20]     -1.17    0.01  0.20   -1.58   -1.29   -1.17   -1.04   -0.75  1411
## alpha[21]     -1.21    0.01  0.22   -1.67   -1.34   -1.20   -1.08   -0.79  1017
## alpha[22]     -1.22    0.01  0.21   -1.66   -1.33   -1.21   -1.09   -0.81  1195
## alpha[23]     -1.00    0.01  0.24   -1.38   -1.16   -1.03   -0.85   -0.47   383
## beta           0.30    0.01  0.17   -0.02    0.19    0.30    0.41    0.64   551
## lp__          -431.17    1.25 12.60 -450.83 -440.39 -433.61 -422.69 -403.34  102
##
##               Rhat
## mu              1.01
## sigma           1.01
## alpha[1]        1.00
## alpha[2]        1.01
## alpha[3]        1.01
## alpha[4]        1.01
## alpha[5]        1.00
## alpha[6]        1.01
## alpha[7]        1.01
## alpha[8]        1.00
## alpha[9]        1.01
## alpha[10]       1.01
## alpha[11]       1.00
## alpha[12]       1.01

```

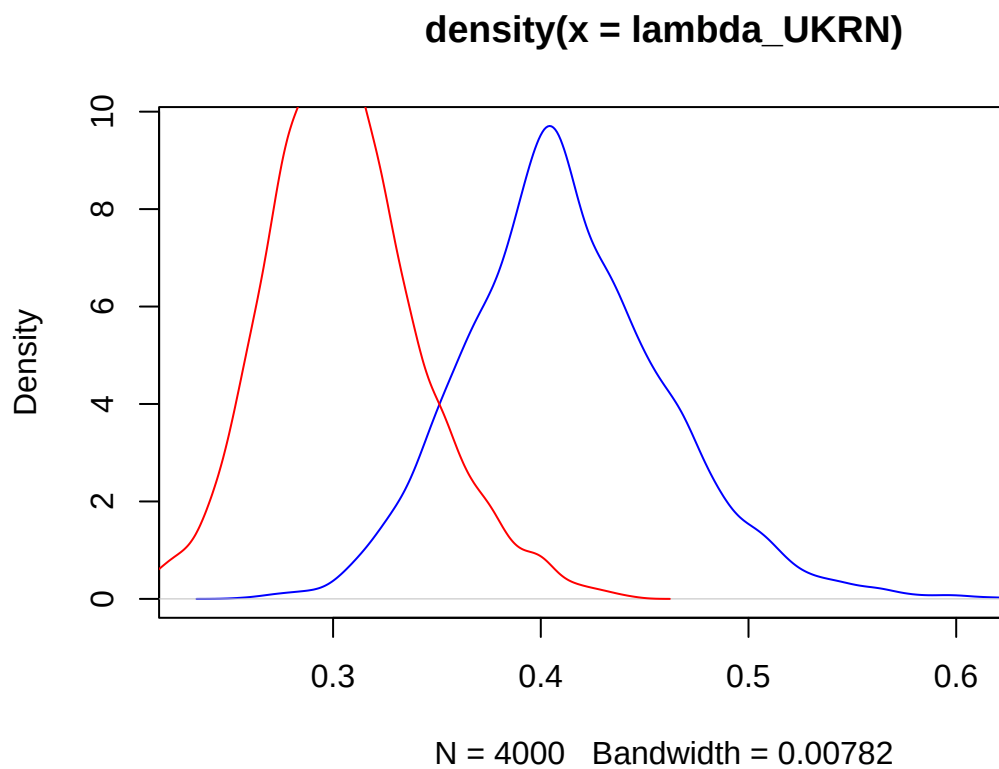
```
## alpha[13] 1.00
## alpha[14] 1.01
## alpha[15] 1.00
## alpha[16] 1.00
## alpha[17] 1.00
## alpha[18] 1.01
## alpha[19] 1.00
## alpha[20] 1.00
## alpha[21] 1.00
## alpha[22] 1.00
## alpha[23] 1.00
## beta      1.01
## lp__      1.03
##
## Samples were drawn using NUTS(diag_e) at Mon Jun 17 12:40:36 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

The output shows that beta credibility intervals do cross 0.

2.5% = -0.10

97.5% = 0.63

So the interpretation is that students who attend a UKRN affiliated university are not more likely to correctly explain open science.



To see the magnitude of this difference.

```
## [1] 0.9675
```

Mean lambda for students in Non-UKRN university = 0.30

Mean lambda for students in a UKRN university = 0.47
Difference = 0.95

Applicability

Bayesian Logistic regression for whether participants think that the issues discussed in the questionnaire are relevant to them. The explainer variables are:

- Whether the participant was at a university that was affiliated with the UKRN - whether the participant was in their first, second, placement of third year

Bayesian model for UKRN/Non-UKRN:

```
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
## Chain 1:
## Chain 1: Gradient evaluation took 0.00033 seconds
## Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 3.3 seconds.
## Chain 1: Adjust your expectations accordingly!
## Chain 1:
## Chain 1:
## Chain 1: Iteration:    1 / 2000 [  0%] (Warmup)
## Chain 1: Iteration:   200 / 2000 [ 10%] (Warmup)
## Chain 1: Iteration:   400 / 2000 [ 20%] (Warmup)
## Chain 1: Iteration:   600 / 2000 [ 30%] (Warmup)
## Chain 1: Iteration:   800 / 2000 [ 40%] (Warmup)
## Chain 1: Iteration:  1000 / 2000 [ 50%] (Warmup)
## Chain 1: Iteration:  1001 / 2000 [ 50%] (Sampling)
## Chain 1: Iteration:  1200 / 2000 [ 60%] (Sampling)
## Chain 1: Iteration:  1400 / 2000 [ 70%] (Sampling)
## Chain 1: Iteration:  1600 / 2000 [ 80%] (Sampling)
## Chain 1: Iteration:  1800 / 2000 [ 90%] (Sampling)
## Chain 1: Iteration:  2000 / 2000 [100%] (Sampling)
## Chain 1:
## Chain 1: Elapsed Time: 3.178 seconds (Warm-up)
## Chain 1:                1.856 seconds (Sampling)
## Chain 1:                5.034 seconds (Total)
## Chain 1:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
## Chain 2:
## Chain 2: Gradient evaluation took 8.4e-05 seconds
## Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.84 seconds.
## Chain 2: Adjust your expectations accordingly!
## Chain 2:
## Chain 2:
## Chain 2: Iteration:    1 / 2000 [  0%] (Warmup)
## Chain 2: Iteration:   200 / 2000 [ 10%] (Warmup)
## Chain 2: Iteration:   400 / 2000 [ 20%] (Warmup)
## Chain 2: Iteration:   600 / 2000 [ 30%] (Warmup)
## Chain 2: Iteration:   800 / 2000 [ 40%] (Warmup)
## Chain 2: Iteration:  1000 / 2000 [ 50%] (Warmup)
## Chain 2: Iteration:  1001 / 2000 [ 50%] (Sampling)
## Chain 2: Iteration:  1200 / 2000 [ 60%] (Sampling)
## Chain 2: Iteration:  1400 / 2000 [ 70%] (Sampling)
```

```

## Chain 2: Iteration: 1600 / 2000 [ 80%] (Sampling)
## Chain 2: Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 2: Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 2:
## Chain 2: Elapsed Time: 1.576 seconds (Warm-up)
## Chain 2: 1.251 seconds (Sampling)
## Chain 2: 2.827 seconds (Total)
## Chain 2:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).
## Chain 3:
## Chain 3: Gradient evaluation took 8.4e-05 seconds
## Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.84 seconds.
## Chain 3: Adjust your expectations accordingly!
## Chain 3:
## Chain 3:
## Chain 3: Iteration: 1 / 2000 [ 0%] (Warmup)
## Chain 3: Iteration: 200 / 2000 [ 10%] (Warmup)
## Chain 3: Iteration: 400 / 2000 [ 20%] (Warmup)
## Chain 3: Iteration: 600 / 2000 [ 30%] (Warmup)
## Chain 3: Iteration: 800 / 2000 [ 40%] (Warmup)
## Chain 3: Iteration: 1000 / 2000 [ 50%] (Warmup)
## Chain 3: Iteration: 1001 / 2000 [ 50%] (Sampling)
## Chain 3: Iteration: 1200 / 2000 [ 60%] (Sampling)
## Chain 3: Iteration: 1400 / 2000 [ 70%] (Sampling)
## Chain 3: Iteration: 1600 / 2000 [ 80%] (Sampling)
## Chain 3: Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 3: Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 3:
## Chain 3: Elapsed Time: 1.893 seconds (Warm-up)
## Chain 3: 1.289 seconds (Sampling)
## Chain 3: 3.182 seconds (Total)
## Chain 3:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).
## Chain 4:
## Chain 4: Gradient evaluation took 9.8e-05 seconds
## Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.98 seconds.
## Chain 4: Adjust your expectations accordingly!
## Chain 4:
## Chain 4:
## Chain 4: Iteration: 1 / 2000 [ 0%] (Warmup)
## Chain 4: Iteration: 200 / 2000 [ 10%] (Warmup)
## Chain 4: Iteration: 400 / 2000 [ 20%] (Warmup)
## Chain 4: Iteration: 600 / 2000 [ 30%] (Warmup)
## Chain 4: Iteration: 800 / 2000 [ 40%] (Warmup)
## Chain 4: Iteration: 1000 / 2000 [ 50%] (Warmup)
## Chain 4: Iteration: 1001 / 2000 [ 50%] (Sampling)
## Chain 4: Iteration: 1200 / 2000 [ 60%] (Sampling)
## Chain 4: Iteration: 1400 / 2000 [ 70%] (Sampling)
## Chain 4: Iteration: 1600 / 2000 [ 80%] (Sampling)
## Chain 4: Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 4: Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 4:

```

```

## Chain 4: Elapsed Time: 1.563 seconds (Warm-up)
## Chain 4: 0.909 seconds (Sampling)
## Chain 4: 2.472 seconds (Total)
## Chain 4:

## Warning: There were 604 divergent transitions after warmup. See
## https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
## to find out why this is a problem and how to eliminate them.

## Warning: There were 3 chains where the estimated Bayesian Fraction of Missing Information was low. See
## https://mc-stan.org/misc/warnings.html#bfmi-low

## Warning: Examine the pairs() plot to diagnose sampling problems

## Warning: The largest R-hat is 1.22, indicating chains have not mixed.
## Running the chains for more iterations may help. See
## https://mc-stan.org/misc/warnings.html#r-hat

## Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and medians may be
## Running the chains for more iterations may help. See
## https://mc-stan.org/misc/warnings.html#bulk-ess

## Warning: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and tail quantiles
## Running the chains for more iterations may help. See
## https://mc-stan.org/misc/warnings.html#tail-ess

## Inference for Stan model: anon_model.
## 4 chains, each with iter=2000; warmup=1000; thin=1;
## post-warmup draws per chain=1000, total post-warmup draws=4000.
##
##               mean se_mean   sd  2.5%  25%   50%   75%  97.5% n_eff
## mu           2.43    0.04  0.20  2.05  2.28  2.42  2.58  2.78    22
## sigma        0.27    0.03  0.18  0.06  0.11  0.24  0.39  0.66    46
## alpha[1]     2.33    0.05  0.32  1.62  2.13  2.34  2.56  2.86    43
## alpha[2]     2.18    0.06  0.36  1.37  1.95  2.22  2.46  2.73    32
## alpha[3]     2.51    0.02  0.34  1.90  2.30  2.48  2.67  3.30   269
## alpha[4]     2.21    0.06  0.36  1.39  2.00  2.25  2.48  2.77    42
## alpha[5]     2.39    0.04  0.33  1.68  2.19  2.40  2.62  2.99    56
## alpha[6]     2.39    0.04  0.33  1.71  2.19  2.40  2.62  3.01    69
## alpha[7]     2.38    0.04  0.32  1.71  2.18  2.40  2.63  3.02    62
## alpha[8]     2.52    0.02  0.33  1.92  2.31  2.50  2.68  3.32   286
## alpha[9]     2.51    0.03  0.33  1.91  2.29  2.48  2.71  3.25    95
## alpha[10]    2.49    0.02  0.32  1.91  2.28  2.48  2.65  3.22   227
## alpha[11]    2.48    0.03  0.31  1.90  2.28  2.48  2.69  3.18   106
## alpha[12]    2.30    0.06  0.33  1.58  2.10  2.32  2.53  2.85    35
## alpha[13]    2.43    0.04  0.30  1.82  2.24  2.42  2.62  3.02    69
## alpha[14]    2.44    0.03  0.31  1.83  2.24  2.43  2.63  3.06   105
## alpha[15]    2.49    0.02  0.31  1.89  2.29  2.48  2.67  3.19   316
## alpha[16]    2.57    0.02  0.33  2.02  2.35  2.54  2.73  3.38   218
## alpha[17]    2.50    0.04  0.33  1.89  2.29  2.48  2.72  3.20    73
## alpha[18]    2.58    0.03  0.35  1.99  2.34  2.54  2.76  3.43   193
## alpha[19]    2.50    0.02  0.32  1.92  2.31  2.49  2.68  3.25   257
## alpha[20]    2.45    0.04  0.33  1.84  2.24  2.44  2.68  3.10    66
## alpha[21]    2.37    0.05  0.31  1.70  2.18  2.37  2.59  2.95    34
## alpha[22]    2.19    0.06  0.36  1.40  1.97  2.23  2.46  2.70    35
## alpha[23]    2.56    0.02  0.33  1.98  2.35  2.54  2.71  3.37   290
## beta        -0.15    0.03  0.25 -0.65 -0.32 -0.16  0.02  0.34    83

```

```

## lp__      -225.07      2.64 14.10 -247.82 -236.35 -227.59 -211.44 -201.46      29
##          Rhat
## mu        1.14
## sigma     1.11
## alpha[1]  1.09
## alpha[2]  1.13
## alpha[3]  1.02
## alpha[4]  1.10
## alpha[5]  1.07
## alpha[6]  1.06
## alpha[7]  1.06
## alpha[8]  1.02
## alpha[9]  1.04
## alpha[10] 1.02
## alpha[11] 1.03
## alpha[12] 1.10
## alpha[13] 1.05
## alpha[14] 1.04
## alpha[15] 1.02
## alpha[16] 1.02
## alpha[17] 1.04
## alpha[18] 1.02
## alpha[19] 1.02
## alpha[20] 1.06
## alpha[21] 1.09
## alpha[22] 1.12
## alpha[23] 1.01
## beta      1.05
## lp__      1.18
##
## Samples were drawn using NUTS(diag_e) at Mon Jun 17 12:40:50 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

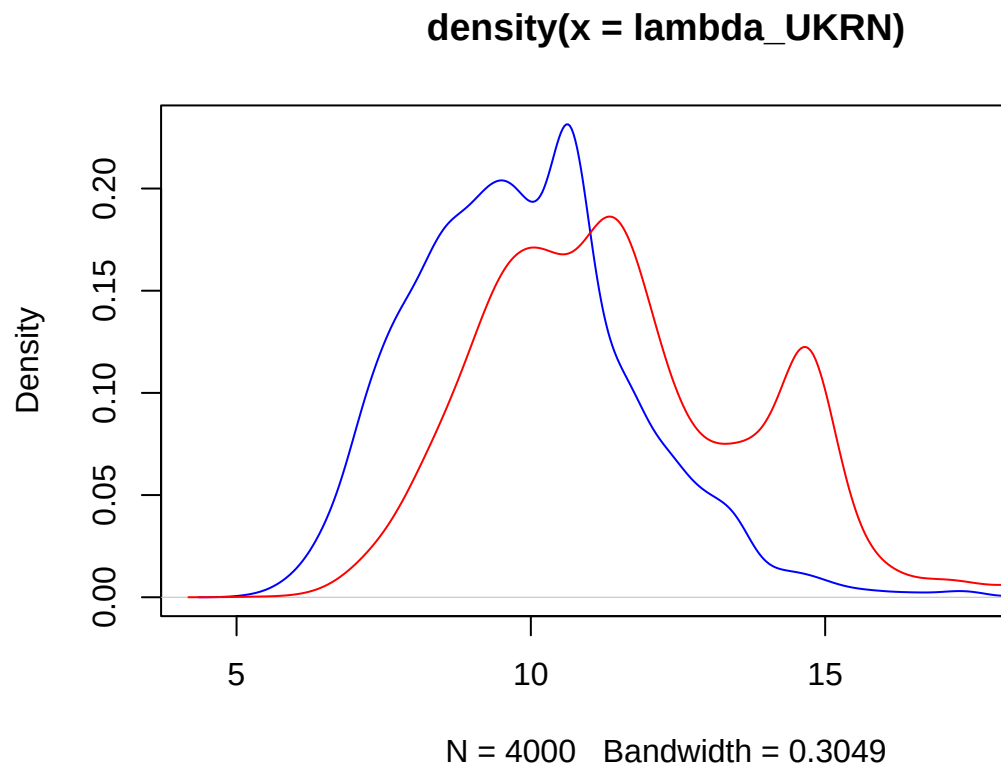
```

The output shows that beta credibility intervals do cross 0.

2.5% = -1.06

97.5% = 0.30

So the interpretation is that students who attend a UKRN affiliated university are not more likely to correctly explain open science.



To see the magnitude of this difference.

```
## [1] 0.27725
```

Mean lambda for students in Non-UKRN university = 13.57

Mean lambda for students in a UKRN university = 13.38

Difference = 0.46